$\textbf{\textit{Regularization Techniques}}$

$Model\ has\ three\ modes$

- $Under\ fit:\ \ High\ bias - Low\ variance$
- $Normal\ fit: Low\ bias - Low\ variance$
- $Overfit\ \ : Low\ bias - High\ variance$

$What\ is\ the\ Reasons\ of\ Overfit$

1) $More\ features$
2) $Less\ data\ set + more\ features$
3) $More\ over\ training (Deep\ learning)$

$In\ order\ to\ avoid\ that\ we\ are\ using\ feature\ selection$

- $p - value$
- $VIF\ (Multicollinearity)$
- $Adjusted\ R_{square}$
- $Information\ gain$
  $This\ methods\ remove\ the\ unwanted\ features$

$There\ is\ a\ need\ I\ dont\ want\ drop\ the\ features\ but\ I\ want\ avoid\ the\ overfit$
$The\ answer\ is\ Regularization\ methods$

1) $Lasso\ Regression$
2) $Ridge\ Regression$

$\boldsymbol{L1\ regularization}$:

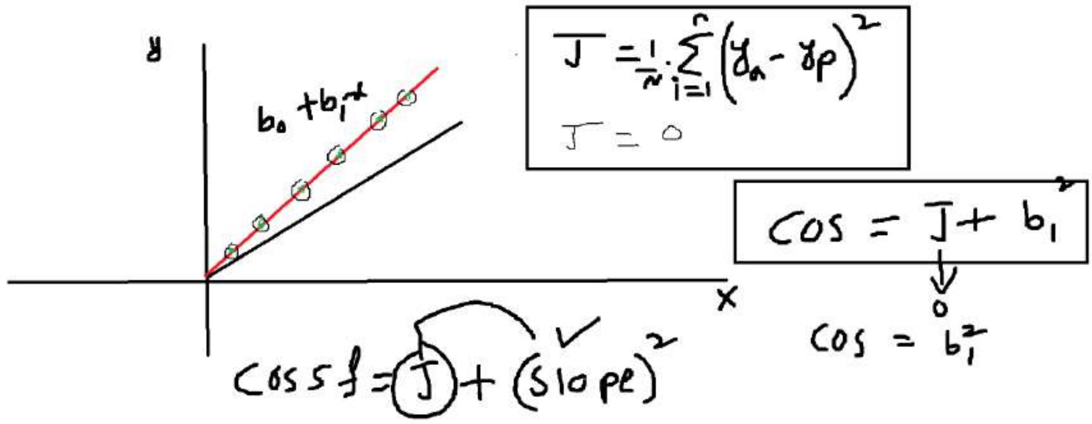
$Power: 1$

$Lasso$

$\boldsymbol{L2Regularization}$:

$Power: 2$

$Ridge$



How Regularization avoids overfitting

- *For any overfitting model , cost function (J) becomes zero*
- *Regulrization methods cost function* $= J + \lambda(slope)^2$ *or* $J + \lambda|Slope|$
- *when we use regulrization methods eventhough J becomes zero, still we can see some error in the form of slope*
- *This indicates still we need to do some developments in order to reduce the cost function*
- *In this process it will automatically avoids the overfitting*

Ridge Regression

1) *In regression the cost function*

$$cost\ function = \frac{1}{n} * \sum_{i}^{n} (y_a - y_p)^2$$

2) *If the model is overfitted then cost function or total error* $= 0$
3) *Regularization techniques avoids the overfitting by modifying the cost*

$function\ equation$

4) $\textbf{Ridge regression}: \ \ \textbf{L2} \ \ \textbf{power}: 2$

$$cost\ function = \frac{1}{n} * \sum_i^n (y_a - y_p)^2 + \ \lambda * (slope)^2$$

$Assume\ that\ the\ regression\ equation:\ b_o + b_1 * x$

- $cost\ function = \frac{1}{n} * \sum_i^n (y_a - y_p)^2 + \lambda * \ (b_1)^2$

$b_o + b_1 * x + b_2 x_2$

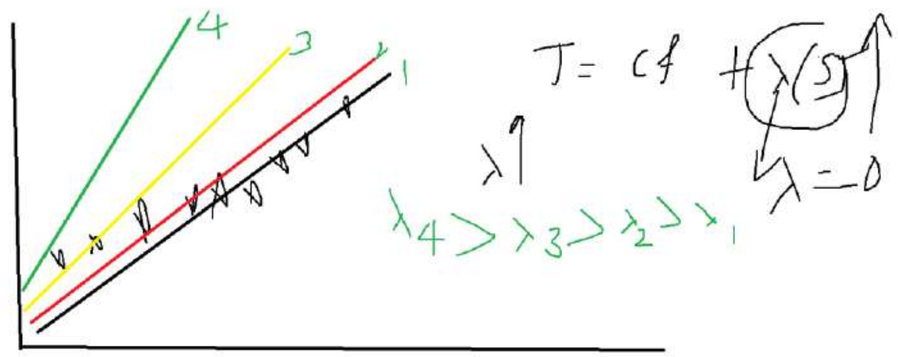
- $cost\ function = Total\ error + \ \lambda * (slope)^2$
- $cost\ function = \frac{1}{n} * \sum_i^n (y_a - y_p)^2 + \ \lambda * [b_1^2 + b_2^2]$

$b_o + b_1 * x + b_2 x_2 + b_3 * x_3$

- $cost\ function = \frac{1}{n} * \sum_i^n (y_a - y_p)^2 + \ \lambda * [b_1^2 + b_2^2 + b_3^2]$

$$cost\ function = \frac{1}{n} * \sum_i^n (y_a - y_p)^2 + \ \lambda * \sum_i^n b_i^2$$

$\lambda = Regularization\ parameter$



$\textbf{How it is working}:$

$Before\ the\ original\ cost\ function\ does\ not\ have\ extra\ term\ slope^2$

- $\frac{1}{n} * \sum_i^n (y_a - y_p)^2$

$When\ the\ model\ will\ be\ overfitted\ automatically\ cost\ function = 0$

- $\frac{1}{n} * \sum_{i}^{n} (y_a - y_p)^2 = 0$

Due to extra $slope^2$ *term, even though total error* $= 0$*, but your cost function*

- $\frac{1}{n} * \sum_{i}^{n} (y_a - y_p)^2 + \lambda * \sum_{i}^{n} b_i^2$

shows still some error in the form of slope

- $cost = 0 + \lambda * \sum_{i}^{n} b_i^2$

we feel such a way that, Still Im seeing the error so I should continue my development

Importance of $\boldsymbol{\lambda}$

λ *= Regularization term*

λ *is called as Hyper parameter, It will provided by user only*

Hyper parameter: The parameters given by user (function arguments)

Parameter: The parameter values given by Model

b_o, b_1 *examples parameters*

$\lambda = 0$

$$Ridge\ cost\ function = \frac{1}{n} * \sum_{i}^{n} (y_a - y_p)^2 + 0 * \sum_{i}^{n} b_i^2$$

Ridge cost function = OLS cost function

λ *= very large*

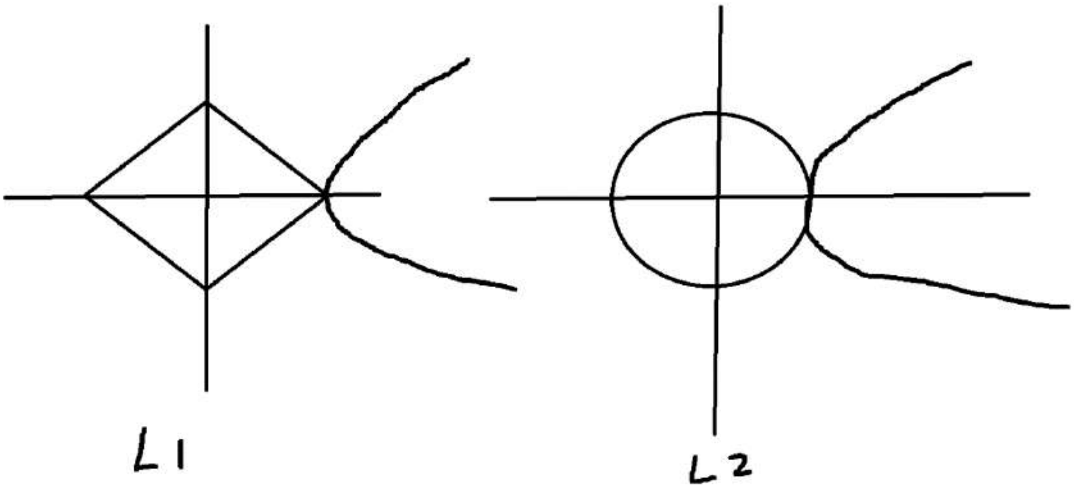
Model wil be underfit

Choose optimal λ *is our hyper parameter*

Lasso Regression: Power = 1

$$cost\ function = \frac{1}{n} * \sum_{i}^{n} (y_a - y_p)^2 + \lambda * \sum_{i}^{n} |b_i|$$

Lasso Vs Ridge



- **Lasso L1 is used as Feature selection**
- **It will used to select only particular columns**
- **Graphical represntation of lasso regression shows less area connected with Cost function that is the indication we can use less features**

Ridge regression is select all the features and avoid the overfitting
Graphical represntation of ridge regression shows as much area connected with Cost function that is the indication we can use all the features

Important points

1) *What is Ridge and what is Lasso Formulae*
2) *One line statement how it avoids overfitting*
3) *What is the importance of* λ
4) $\lambda = 0$
5) $\lambda = max$
6) *when we will use Ridge and when we will use lasso*

IF WE USE BOTH METHODS TOGETHER THAT IS CALLED AS

ELASTIC REGRESSION

$In\ order\ to\ avoid\ that\ we\ are\ using\ feature\ selection$

- $p - value$
- VIF
- $Adjusted\ R_{square}$
- $LASSO$